

Investigating Individual Differences in Olfactory Adaptation to Pulse Ejection Odor Display by Scaling Olfaction Sensitivity of Intensity

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ABSTRACT

Olfactory adaptation is a non-negligible issue to consider to provide sustained olfaction experience in virtual reality. This study conducted experiments to measure users' adaptation to pulse ejection odor display by ink-jet devices and analyzed the individual differences in olfactory perception. The results revealed that the average intensity perception dropped approximately to 70% of max perceived intensity in the 10-minute scent display session. Furthermore, individuals' adaptation levels were correlated to personal sensitivity to olfactory intensity variations acquired by labeled magnitude scale. This work provided a theoretical basis to personalize odor display in VR for more stable olfactory experience.

Index Terms: Human-centered computing—Interaction design—Interaction design theory, concepts and paradigms; Human-centered computing—Human computer interaction (HCI)—Human-centered computing—HCI design and evaluation methods User studies

1 INTRODUCTION

In recent years, odor display has become a vital element in enhancing multi-sensory immersive experience in virtual reality (VR). However, olfactory adaptation is an issue necessary to consider for digital odor display to provide stable and sustained olfaction experience, which is a mechanism that olfactory perception is weakened after prolonged exposure to a certain kind of odor. A new olfactory display method using ink-jet type devices has been developed, which uses directional micro airflow to eject odorant substance to the user's nose [3]. The application of this kind of devices in VR is often in the form of pulse ejection, which ejects scents for a short duration synchronized with respiration. Adaptation to pulse ejection odor display remains unclear since its olfaction experience is quite different from usual scent diffusers that create ambient and continuous scents. Although previous studies have discovered that the method of pulse ejection has the effect of preventing olfactory adaptation for approximately 2 minutes [3], there still lacks quantitative intensity perception evaluation in the time domain and individual differences analysis. Concerning the possible causes of individual differences in olfactory perception and adaptation, we raise a hypothesis that the level of olfactory adaptation is related to personal perceptual sensitivity to the odor intensity.

In this study, we will focus on two objectives regarding the pulse ejection odor display method by ink-jet devices. The first is to evaluate the perceived olfactory intensity with high scaling resolution and time resolution during a reasonable period of time for VR odor display and scale the degree of olfactory adaptation to quantity. The second objective is to investigate users' personal sensitivities to



Figure 1: The experiment environment and devices.

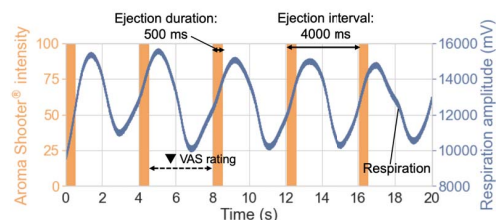


Figure 2: The timing and intensity of pulse ejections. The blue line indicates respiration data acquired using Biosignalsplux® in the designing process.

olfactory intensity and their relationship with olfactory adaptation metrics to verify our hypothesis.

2 EXPERIMENT AND EVALUATION

We conducted a user experiment where participants quantitatively rated their perceived intensity of presented pulse ejection of scents. 11 healthy participants (8 males and 3 females) at the average age of 24.36 ± 1.43 took part in the experiment. The participants were confirmed to not have any sensation disorders.

In this experiment, we used an ink-jet scent display device called Aroma Shooter® (Aromajoin) (Fig. 1c) for pulse ejection display. It provides functions of temporal control over ejection duration and intensity control of 100 linear phases. In this experiment, we used the scent of orange, which is reported pleasing [1], without causing negative emotions that may influence participants' judgment.

For scent display method, we adopted the pulse ejection method in Kadowaki et al.'s work [3] with some adjustments. The duration of each emission was set to 500 ms, and the pace of ejection is synchronized with the respiration rate (15 times per minute). Participants were asked to regulate their respiration pace following audio guidance to exclude the influence of different respiration rates. The timing of each ejection was controlled to be in the effective inhalation range of each respiration, as shown in Fig. 2. The display intensity of Aroma Shooter® was set to 100.

The procedure of the experiment included 4 steps—reference display, the resting session, posture adjustment, and intensity rating of scent display. Before the resting session, participants were displayed with one time of scent emission as the reference to the following intensity rating to ensure the absolute values of intensity ratings were at the same standard across participants. Then, after a 2-minute

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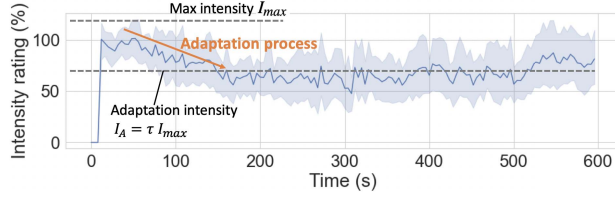


Figure 3: The average perceived intensity results (N=11) during 10 minutes of pulse ejection scent display. The max intensity and adaptation intensity for adaptation score calculation are noted in the graph.

resting session, participants were asked to fix their head position 20 cm away from the Aroma Shooter[®], facing directly to the emission portal as shown in Fig. 1a. In the intensity rating session (10 minutes), after each pulse ejection was perceived and before the next, participants were asked to record their rating of relatively perceived intensity based on the previous reference on the visual analogue scale (VAS) bar as shown in Fig. 1b. The scaling resolution was 1% and the range was from 0% to 200%. During both the resting and the rating sessions, participants were asked to regulate their respiration following an audio guidance. We controlled the air circulation in the room so that there was no perceivable airflow that could interfere the emission direction of Aroma Shooter[®].

To quantify the measurement of olfactory adaptation for statistical analysis, we calculated an adaptation score from the results of perceived intensity rating. First, we defined an adaptation threshold τ in the range of $[0, 1]$ to define whether adaptation occurred or not relative to the maximum perceived intensity I_{max} during odor display. Then we calculated the threshold intensity of adaptation $I_A = \tau I_{max}$ (Fig. 3) to acquire the adaptation time T_A by Equation 1, which was the total amount of time when perceived intensity was below the adaptation intensity.

$$T_A = \sum_{I_t < I_A} t \quad (1)$$

where I_t was the perceived intensity on the time point t . Finally, the adaptation score was acquired by Equation 2, which was the fraction of adaptation time to the total amount of time T of display duration. Therefore, the adaptation score lies in the range of $[0, 1]$, where 0 means that adaptation of the threshold does not occur at all and 1 vice versa.

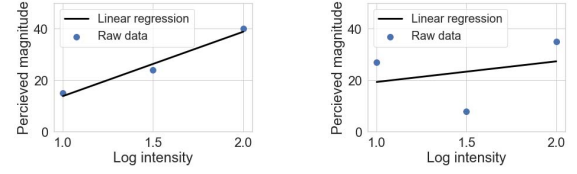
$$S_{Adaptation} = \frac{T_A}{T} \quad (2)$$

We proposed to perform labeled magnitude scale (LMS) to determine individual perceptual sensitivity to odor intensity. LMS is a semantic scale of perceptual intensity characterized by a quasi-logarithmic spacing of its verbal labels (e.g. “weak”, “strong”) [2]. We set 3 values of logarithmic odor intensity presented to participants to determine their intensity perception. The logarithmic values of Aroma Shooter[®] intensities were 1 ($= \log_{10} 10$), 1.5 ($\approx \log_{10} 63$), and 2 ($= \log_{10} 100$). The LMS evaluation was conducted before or after the intensity rating experiment, and the interval between them was random but long enough to make sure that the participant’s olfactory sensation was fully restored.

3 RESULTS AND DISCUSSION

The average results (N = 11) of perceived intensity during the scent display session are shown in Fig. 3. In general, the perceived intensity was in a decreasing phase from around 100% to 70% of the max intensity approximately in the first 180 seconds of the display. Afterward, the perceived intensity tended to stay and fluctuate in the range of around 60% to 80% of the max perceived intensity.

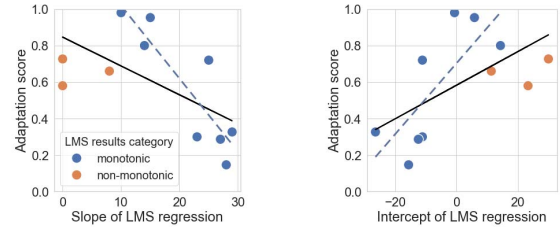
Regarding LMS evaluation results, we performed a linear regression on the LMS results for each participant. Two features of linear regression were considered—the slope and the intercept. The slope



(a) Monotonic LMS result (P4).

(b) Non-monotonic LMS result (P9).

Figure 4: Examples of monotonic and non-monotonic results of LMS evaluation.



(a) The distribution of adaptation scores and slope values of LMS regression.

(b) The distribution of adaptation scores and intercept values of LMS regression.

Figure 5: The distribution of adaptation scores ($\tau=0.8$) and LMS regression features (N=11). The blue and orange dots represent participants with monotonic and non-monotonic LMS results. The black line is the linear regression of all participants’ data, and the blue dashed line is the linear regression of the LMS monotonic group.

of LMS linear regression reflects participants’ sensitivity to the variations of intensities and the intercept reflects the general perception level of the same displayed olfactory intensity. We observed that among 11 participants, there were two categories of results regarding whether the perceived intensity scale and the displayed olfactory intensity were monotonically increasing, as shown in Fig. 4. Participants with monotonic LMS results (Fig. 4a) are considered to have the ability to perceive the relative intensities of odor correctly, and those whose LMS results were not monotonic (Fig. 4b) are considered insensitive to olfactory intensity differences. We examined the Pearson correlations and discovered that the adaptation scores were negatively correlated to the slope values of LMS linear regression ($r = 0.60$), and positively correlated to the intercept values ($r = 0.57$), especially for τ value of 0.8 (Fig. 5). When non-monotonic LMS data was excluded, the correlation appeared to be stronger ($r = -0.87$ for correlation with slope values and $r = 0.75$ for the intercepts). The correlation indicated that different perceptual sensitivity to olfactory intensity variations and the general sensation intensity level of users might be some of the possible causes of individual differences in olfactory adaptation to pulse ejection of odors. The discovery provided us a future direction of personalizing odor intensity according to individual olfactory sensitivities.

Future work to solve the limitations of this study will be increasing participant numbers, verifying the correlation for other scents, and adding comparison condition of odor display to verify the authenticity of the adaptation metric.

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